

Module 1

HARMONIC OSCILLATIONS

PERIODIC MOTION

Any motion which repeats itself at regular intervals of time.

Eg:1. Oscillation of simple pendulum

2.The motion of earth around the sun

3. Oscillations of a loaded spring

OSCILLATORY MOTION

If the particle moves back and forth along the same path and repeats itself at regular intervals of time.

Eg.1. osc. of simple pendulum and loaded spring

2. vibrations of tuning fork.

Here the displacement can be represented by sine and cosine functions or their combinations. Sine and cosine functions are called harmonic functions.

Frequency (f) : No. of oscillations per sec.

Time period (T) : Time required for one oscillation

$$T = 1/f$$

Displacement (x) :The distance of the particle from the equilibrium position.

Amplitude (A) : Maximum displacement

SIMPLE HARMONIC MOTION (SHM)

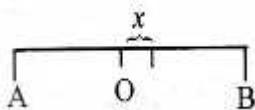
A particle is said to execute SHM if it moves on either side of a fixed point such that it has an acceleration which is directly proportional to the displacement from the fixed point and always directed towards that point.

Eg.1) Oscillations of a simple pendulum

2) electromagnetic oscillations in an LCR circuit.

A body moving simple harmonically is called a harmonic oscillator

Equation of SHM



Consider a particle of mass 'm' executing SHM along a straight line. 'O' is the equilibrium position. Let 'x' be the displacement at an instant.

$$\text{Acceleration of the particle} = \frac{d^2x}{dt^2}$$

$$\text{Restoring force on the particle} = ma = m \frac{d^2x}{dt^2}$$

$$\text{By definition of SHM, } m \frac{d^2x}{dt^2} \propto -x$$

$$m \frac{d^2x}{dt^2} = -kx$$

Where 'k' is the proportionality constant. Negative sign indicates that restoring force acts against displacement.

$$m \frac{d^2x}{dt^2} + kx = 0$$

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

$$\text{put } \frac{k}{m} = \omega^2$$

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \quad \text{--- (1)}$$

This is the differential equation for SHM.

Solution of the equation of motion

$$\text{We have } \frac{d^2x}{dt^2} + \omega^2 x = 0$$

$$\text{Multiplying the equation by } 2 \frac{dx}{dt},$$

$$2 \frac{dx}{dt} \cdot \frac{d^2x}{dt^2} + 2 \frac{dx}{dt} \omega^2 x \text{ --- (2)}$$

Eq(2) can be written as

$$\frac{d}{dt} \left[\left(\frac{dx}{dt} \right)^2 + \omega^2 x^2 \right] = 0$$

$$\text{Integrating, } \left(\frac{dx}{dt} \right)^2 + \omega^2 x^2 = C \text{ --- (3)}$$

Where C is the constant of integration.

To find C:

The velocity of the particle at the extreme position is zero. If 'a' is the amplitude, Then

$$\frac{dx}{dt} = 0, \text{ at } x = a$$

putting in equation (3) we get $C = \omega^2 a^2$

putting this in eq (3)

$$\left(\frac{dx}{dt} \right)^2 + \omega^2 x^2 = \omega^2 a^2$$

$$\left(\frac{dx}{dt} \right)^2 = \omega^2 (a^2 - x^2)$$

$$\frac{dx}{dt} = \omega \sqrt{(a^2 - x^2)} \quad \text{--- (4)}$$

$$\text{ie velocity} = \omega \sqrt{(a^2 - x^2)} \quad \text{--- (5)}$$

$$\text{From eq (4), } \frac{dx}{\sqrt{(a^2 - x^2)}} = \omega dt$$

$$\text{Integrating, } \sin^{-1} \frac{x}{a} = \omega t + \phi$$

where ϕ is constant of integration.

$$\text{ie } \frac{x}{a} = \sin(\omega t + \phi)$$

$$x = a \sin(\omega t + \phi) \quad \text{--- (6)}$$

$(\omega t + \phi)$ is the phase of oscillation at any instant.

$$\text{let initial phase } \phi = \delta + \frac{\pi}{2}$$

$$\text{Then } x = a \sin \left(\omega t + \delta + \frac{\pi}{2} \right)$$

$$x = a \cos (\omega t + \delta) \text{ --- (7)}$$

this also represents SHM .

If time t in eq (6) is increased by $\frac{2\pi}{\omega}$,

$$x = a \sin \left[\omega \left(t + \frac{2\pi}{\omega} \right) + \phi \right] =$$

$$x = a \sin (\omega t + 2\pi + \phi)$$

$$x = a \sin (\omega t + \phi)$$

The function repeats itself after a time $\frac{2\pi}{\omega}, \frac{4\pi}{\omega}$ etc .

Hence it is called Period $T \therefore T = \frac{2\pi}{\omega}$

$$\text{But } \omega^2 = \frac{k}{m}; \quad \omega = \sqrt{\frac{k}{m}} \quad \therefore T = 2\pi \sqrt{\frac{m}{k}}$$

FREE OSCILLATIONS

Consider a particle executing oscillations. If no forces are acting on it, the oscillations will continue for an indefinite period without change in amplitude. In this oscillation total energy of the system remains the same always. This type of oscillations is called free oscillations

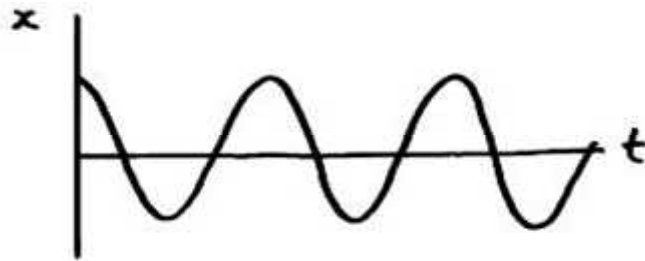


Fig: Free Oscillations

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$$

DAMPED HARMONIC OSCILLATION

A harmonic oscillator in which the motion is damped by the action of an additional force is called a damped harmonic oscillator. Damped oscillations are oscillations under the action of resistive forces.

Damping force is $\propto$ to its velocity.

$$\text{ie Damping force} = b \frac{dx}{dt}$$

'b' is called the damping constant. Including the damping force, the differential eq. for a damped harmonic oscillator is

$$m \frac{d^2 x}{dt^2} = -kx - b \frac{dx}{dt}$$

$$m \frac{d^2 x}{dt^2} + kx + b \frac{dx}{dt} = 0$$

$$\frac{d^2 x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x = 0 \text{ --- (1)}$$

Let $\frac{b}{m} = 2\gamma$ where γ is the damping coefft .

$$\frac{d^2 x}{dt^2} + 2\gamma \frac{dx}{dt} + \omega^2 x = 0 \text{ --- (2)}$$

This is the differentti al eqn for damped harmonic oscillator

$\sqrt{\frac{k}{m}} = \omega_0$ = natural angular frequency of oscillator in the absence of damping force.

Solutions to the Equation

Assume the solution is of the form

$$x = Ae^{\alpha t}$$

$$\frac{dx}{dt} = \alpha Ae^{\alpha t}$$

$$\frac{d^2 x}{dt^2} = \alpha^2 Ae^{\alpha t} = \alpha^2 x$$

Putting these values in (2), we get

$$\alpha^2 x + 2\gamma \alpha x + \omega^2 x = 0$$

$$\text{ie } \alpha^2 + 2\gamma \alpha + \omega^2 = 0$$

$$\text{The solution } \alpha = \frac{-2\gamma \pm \sqrt{(4\gamma^2 - 4\omega^2)}}{2}$$

$$x = Ae^{(-\gamma \pm \sqrt{\gamma^2 - \omega^2})t}$$

The solutions can be written as

$$x_1 = A_1 e^{(-\gamma + \sqrt{\gamma^2 - \omega^2})t} \text{ and } x_2 = A_2 e^{(-\gamma - \sqrt{\gamma^2 - \omega^2})t}$$

$\therefore$ The general solution is $x = x_1 + x_2$

$$x = A_1 e^{(-\gamma + \sqrt{\gamma^2 - \omega^2})t} + A_2 e^{(-\gamma - \sqrt{\gamma^2 - \omega^2})t} \text{ --- (3)}$$

Where A_1 and A_2 are constants which depends on the initial values of position and velocity .
' γ ' determines the behaviour of the system .

Case 1: $\gamma > \omega$

The roots of the auxiliary equation are real

$$\alpha_1 = -\gamma + \sqrt{(\gamma^2 - \omega^2)}; \quad \alpha_2 = -\gamma - \sqrt{(\gamma^2 - \omega^2)} \quad \dots (4)$$

The solution of equation (3) is

$$x = A_1 e^{\alpha_1 t} + A_2 e^{\alpha_2 t} \quad \dots (5)$$

Where A_1 and A_2 are constants.

Let the mass be given a displacement x_0 and then released, So that

$$x = x_0, \quad \frac{dx}{dt} = 0 \quad \text{at} \quad t = 0$$

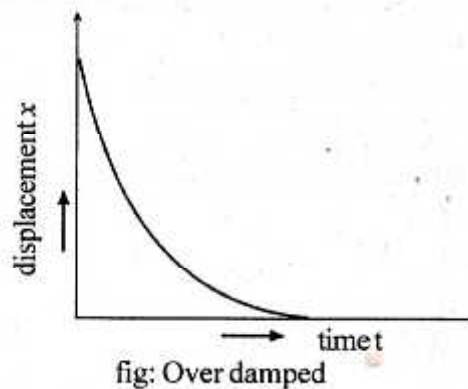
$$\text{From eq (5) we get} \quad x_0 = A_1 + A_2$$

$$\text{and} \quad A_1 \alpha_1 + A_2 \alpha_2 = 0$$

$$\text{Simplifyin g, } A_1 = -\frac{x_0 \alpha_2}{\alpha_1 - \alpha_2} \quad \text{and} \quad A_2 = \frac{x_0 \alpha_1}{\alpha_1 - \alpha_2} \quad \dots (6)$$

$$\therefore \text{Eq (5) reduces to} \quad x = \frac{x_0}{\alpha_1 - \alpha_2} (\alpha_1 e^{\alpha_2 t} - \alpha_2 e^{\alpha_1 t})$$

This shows that x is always positive and decreases exponentially to zero without any oscillations. The motion is called over damped or dead beat motion



Case 2: $\gamma = \omega$

The roots of the auxiliary equation are real and equal each being equal to $-\gamma$.

The general solution of eq (2) is then

$$x = (A_1 + A_2 t) e^{-\gamma t}$$

Here also if $x = x_0, \quad \frac{dx}{dt} = 0 \quad \text{at} \quad t = 0$

The constants $A_1 = x_0$ and $A_2 = \gamma x_0$

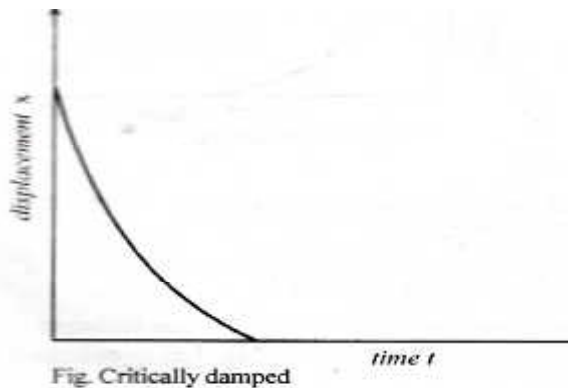
Hence the solution to the equation (2) is

$$x = x_0 (1 + \gamma t) e^{-\gamma t}$$

Again x is always positive and decreases to zero. The nature of the motion is similar to that of case 1. But it is different from the non-oscillatory motion of case 1. Hence this motion is termed as critically damped. Here the particle acquires the position of equilibrium very rapidly.

Application

It is applied in automobile shock absorbers door closer mechanism and recoil mechanism in gun. If the door closes without performing any oscillation, it is an example of critically damped system.



Case3: $\gamma < \omega$ The damping is low

Here $\sqrt{\gamma^2 - \omega^2}$ is imaginary

Let $\sqrt{\gamma^2 - \omega^2} = i\omega' = i\sqrt{\omega^2 - \gamma^2}$

Where $\omega' = \sqrt{\omega^2 - \gamma^2}$ -----(8)

From eq (3),

$$x = A_1 e^{(-\gamma + i\omega')t} + A_2 e^{(-\gamma - i\omega')t} \text{ -----(9)}$$

$$= e^{-\gamma t} (A_1 e^{i\omega' t} + A_2 e^{-i\omega' t})$$

$$= e^{-\gamma t} (A_1 (\cos \omega' t + i \sin \omega' t)) + A_2 (\cos \omega' t -$$

$$i \sin \omega' t)$$

$$X = e^{-\gamma t} ((A_1 + A_2) \cos \omega' t + i(A_1 - A_2) \sin \omega' t)$$

$$\text{Put } (A_1 + A_2) = x_0 \sin \varphi$$

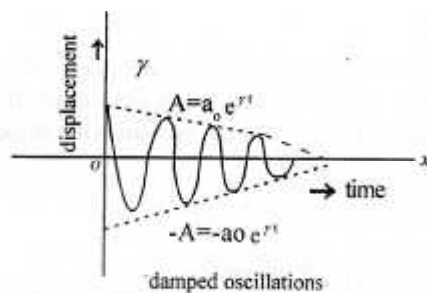
$$\text{And } i(A_1 - A_2) = x_0 \cos \varphi$$

$$X = x_0 e^{-\gamma t} \sin(\omega' t + \varphi) \text{ --- (10)}$$

Equation (10) shows that the motion is oscillatory. The amplitude $x_0 e^{-\gamma t}$ is not a constant but decreases with time. The motion is not periodic.

$$\text{Period of oscillation } T = \frac{2\pi}{\omega'} = \frac{2\pi}{\sqrt{\omega^2 - \gamma^2}}$$

The oscillations are as shown in the figure.



Effect of damping

1. Amplitude of oscillations decreases exponentially with time.
2. The frequency of oscillation of a damped oscillator is less than the frequency of undamped oscillation. i.e. Frequency of damped oscillation is reduced from the natural frequency ω_0 .
3. The energy of the damped oscillator decreases exponentially with time.
4. Energy is in the form of Kinetic and Potential. A part of the stored energy is dissipated in every cycle.

The average energy over a cycle is given by

$$E = E_0 e^{-2\gamma t}$$

$$= E_0 e^{-\frac{t}{\tau}} \text{ where } \tau = \frac{1}{2\gamma} = \frac{m}{b} \text{ is called the relaxation time } (\tau)$$

This shows that the energy stored in the system decreases exponentially with time for every cycle. E_0 is the initial energy and E is the average energy per cycle.

When $t = \tau$, $E = \frac{E_0}{e}$ (where e is the base of natural logarithm).

$$e=2.72$$

Hence the relaxation time of an oscillating system may be defined as the time taken by the system to reduce its average total energy to $1/e$ (37%) of its initial energy

FORCED HARMONIC OSCILLATOR

If an external force acts on a damped oscillatory system it is called a Forced Harmonic Oscillator. The oscillations produced, under the action of external periodic force on the body is called forced oscillations.

The energy of a damped oscillator decreases with the passage of time. It is possible to compensate for the energy loss by applying a suitable external periodic force. For eg: a swing.

The frequency of the forced oscillation will be different from the natural frequency of the body. Let ω be the frequency of natural and ω_f the frequency of the forced vibrations when the body is subjected to an oscillatory external force $F_0 \sin \omega_f t$.

The Equation of motion is given by

$$m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = F_0 \sin \omega_f t$$

$$\frac{d^2 x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x = \frac{F_0}{m} \sin \omega_f t \text{-----(1)}$$

Let $\frac{b}{m} = 2\gamma$ where γ is the damping coefft and $\omega^2 = \frac{k}{m}$

$$\frac{d^2 x}{dt^2} + 2\gamma \frac{dx}{dt} + \omega^2 x = f_0 \sin \omega_f t \text{-----(2)}$$

where $f_0 = \frac{F_0}{m}$

This is the differential eqn for damped harmonic oscillator

Here a tussle occurs between the external periodic force and the damping force. After some time the system reaches a steady state. The resulting oscillations are forced oscillations. Its frequency is that of the external periodic force and not that of the natural frequency of the body. When steady state is reached, the solution of the above equation is given by

$$x = A \sin(\omega_f t - \theta) \text{-----(3)}$$

$$\frac{dx}{dt} = A \omega_f \cos(\omega_f t - \theta)$$

$$\frac{d^2 x}{dt^2} = -A \omega_f^2 \sin(\omega_f t - \theta)$$

Putting these values in (2)

$$-A \omega_f^2 \sin(\omega_f t - \theta) + 2\gamma A \omega_f \cos(\omega_f t - \theta) + \omega^2 A \sin(\omega_f t - \theta) = f_0 \sin[(\omega_f t - \theta) + \theta] \text{-----(4)}$$

Expanding RHS

$$= f_0 \sin(\omega_f t - \theta) \cos \theta + f_0 \cos(\omega_f t - \theta) \sin \theta$$

$$\begin{aligned} & \left[-A\omega_f^2 - f_0 \cos \theta + \omega^2 A \right] \sin(\omega_f t - \theta) + \\ & \left[2\gamma A \omega_f - f_0 \sin \theta \right] \cos(\omega_f t - \theta) = 0 \quad \text{--- (5)} \end{aligned}$$

To find A :

For this eq. to hold good for all values of t , the coeffs of the term $\sin(\omega_f t - \theta)$ and $\cos(\omega_f t - \theta)$ must vanish separately.

$$\text{ie } \left[-A\omega_f^2 - f_0 \cos \theta + \omega^2 A \right] = 0$$

$$\therefore -A\omega_f^2 + \omega^2 A = f_0 \cos \theta \quad \text{--- (6)}$$

$$\left[2\gamma A \omega_f - f_0 \sin \theta \right] = 0$$

$$\therefore \left[2\gamma A \omega_f = f_0 \sin \theta \right] \quad \text{--- (7)}$$

Squaring and eq. (6) and (7),

$$\left(-A\omega_f^2 + \omega^2 A \right)^2 + 4\gamma^2 A^2 \omega_f^2 = f_0^2$$

$$A = \frac{f_0}{\sqrt{(\omega^2 - \omega_f^2)^2 + 4\gamma^2 \omega_f^2}} \quad \text{--- (8)}$$

This is the amplitude of the Forced vibration.

Dividing eq. (7) by (6)

$$\tan \theta = \frac{2\gamma \omega_f}{\omega^2 - \omega_f^2} \quad \text{--- (9)}$$

This gives the phase difference between the forced oscillations and the applied force.

Substituting for A in eq. (3)

$$x = \frac{f_0 \sin(\omega_f t - \theta)}{\sqrt{(\omega^2 - \omega_f^2)^2 + 4\gamma^2 \omega_f^2}}$$

The above equation shows that the system vibrates with the frequency of the applied periodic force and is having a phase lag of θ .

Different cases :

Case (1) $\omega_f < \omega$ (low driving frequency)

$$\text{We have } A = \frac{f_0}{\sqrt{(\omega^2 - \omega_f^2)^2 + 4\gamma^2 \omega_f^2}}$$

$$\text{Neglecting } \omega_f^2, \quad A = \frac{f_0}{\omega^2} = \frac{\frac{F_0}{m}}{\frac{k}{m}} = \frac{F_0}{k}$$

Amplitude does not depend on the mass of the oscillating body, but depends on the force constant.

case 2 : $\omega_f = \omega$ (Resonance condition)

The particular frequency of the applied periodic force at which the amplitude becomes maximum is called resonant frequency. This will happen when the frequency of the periodic force becomes equal to the natural frequency of the oscillator .

Amplitude at Resonance $A_{\max} = \frac{f_0}{2\gamma\omega_f}$

$$\tan \theta = \frac{2\gamma\omega_f}{0} = \infty ; \quad \theta = \frac{\pi}{2}$$

The displacement lags the applied voltage by $\frac{\pi}{2}$.

The value of amplitude depends upon the damping coefft, γ .

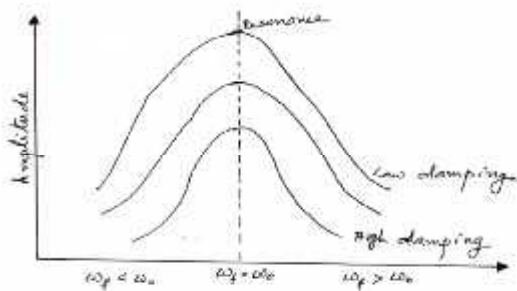
Actually resonance occurs at a frequency slightly less than ω .

Case 3 : $\omega_f > \omega$ (high driving frequency)

$$A = \frac{f_0}{\sqrt{(\omega^2 - \omega_f^2)^2 + 4\gamma^2\omega_f^2}}$$

When $\omega_f > \omega$ and for low damping $A = \frac{f_0}{\omega_f^2}$

Graph: Variation of Amplitude A with ω



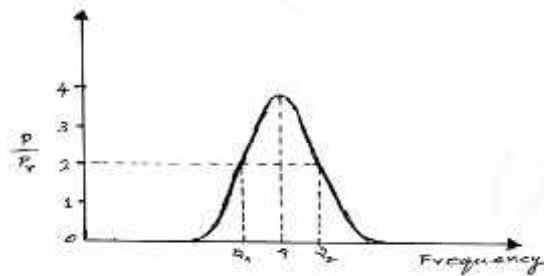
Sharpness of Resonance

When ω_f is increased or decreased from resonance frequency, the amplitude falls from the maximum value.

The term sharpness of resonance refers to the rate of decrease of amplitude with the change in frequency of the applied periodic force on either side of the resonant frequency.

Let P_r be the power absorbed at resonance and P , the power absorbed at any other frequency γ .

The graph between P/P_r and frequency is as follows



At the frequency values the power absorbed is half of the maximum and these γ_1 and γ_2 values are called half power points. The frequency difference between these two half power points is called the bandwidth of the oscillator.

ie Bandwidth $\Delta \gamma = \gamma_2 - \gamma_1$

Band width is a range of frequencies for which the power is greater than or equal to half of the power at resonant frequency.

Sharpness of resonance is defined as the ratio of resonant frequency to bandwidth

$$\begin{aligned}\text{Sharpness of resonance} &= \gamma / \gamma_2 - \gamma_1 \\ &= \gamma / \Delta \gamma\end{aligned}$$

A very sharp resonant circuit has a very small bandwidth

QUALITY FACTOR

Quality factor represents the efficiency of the oscillator. It is a dimensionless parameter that describes the oscillations of damped oscillator. High value of Q indicates low rate of energy loss.

Q - factor is defined as 2π times the energy stored in the oscillator to the energy dissipated per cycle.

ie

$$Q = \frac{2\pi \times \text{Energy stored in the oscillator}}{\text{Energy lost per cycle}}$$

$$= \frac{2\pi \times \text{Energy stored in the oscillator}}{\text{Energy lost per sec} \times \text{Time period}}$$

$$= \frac{2\pi E}{-\frac{dE}{dt} T} \text{ --- (1) where } E \text{ is the energy. } -\frac{dE}{dt} \text{ is the power}$$

dissipated per sec, P.

Energy of a damped harmonic oscillator is given by $E = E_0 e^{-2\gamma t}$

where E_0 is the initial energy.

$$\text{But Power } P = \frac{dE}{dt} = -2\gamma E_0 e^{-2\gamma t}$$

$$\text{But Power dissipated } P = -\frac{dE}{dt} = 2\gamma E$$

$$Q = \frac{2\pi E}{PT} = \frac{2\pi E}{2\gamma E T} = \frac{\pi}{\gamma T} = \frac{\pi}{\gamma \frac{2\pi}{\omega_0}} = \frac{\omega_0}{2\gamma}$$

$$\therefore Q = \frac{\omega_0}{2\gamma}. \text{ But } \frac{1}{2\gamma} = \tau, \text{ the relaxation time.}$$

$$\therefore Q = \omega_0 \tau$$

$$\frac{Q}{2\pi} = \frac{\omega_0 \tau}{2\pi} = f_0 \tau$$

Where frequency f_0 gives the number of oscillations in one second.

Hence $f_0 \tau$ gives the number of oscillations during the relaxation time
Q will be large if γ is small and Q will be small if γ is large

LCR circuit as an Electrical analogue of mechanical oscillator

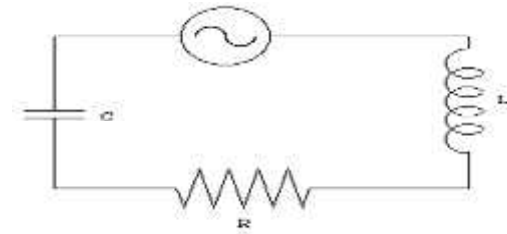
A pure LC circuit is an electrical analogue of simple pendulum. In the case of simple pendulum, the energy alternates between potential and kinetic energy.

In the case of LC circuit when the electrical oscillation takes place, the energy is alternately shared in the capacitor as electric field and in the inductor as magnetic field.

The frequency oscillation in LC circuit, $n = \frac{1}{2\pi\sqrt{LC}}$

Forced oscillation in a series LCR circuit

The sum of the voltage drops across the 3 components must be equal to the applied voltage



Applying Kirchoff 's law to the circuit

$$V_L + iR + V_C = V_0 \sin \omega t$$

$$L \frac{di}{dt} + iR + V_C = V_0 \sin \omega t$$

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = V_0 \sin \omega t$$

$$\text{ie } \frac{d^2 q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{q}{LC} = \frac{V_0}{L} \sin \omega t$$

This is the differential equation in the case of forced oscillations in LCR circuit

COMPARISON between Mechanical and Electrical Oscillator

Mechanical Oscillator $\rightarrow \frac{d^2 x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x = \frac{F_0}{m} \sin \omega_f t$

Electrical Oscillator $\rightarrow \frac{d^2 q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{q}{LC} = \frac{V_0}{L} \sin \omega t$

Mechanical Oscillator

1. Displacement x
2. Velocity dx/dt
3. Mass m
4. Damping coefficient γ
5. Force amplitude f_0
6. Driving frequency ω_f

Electrical Oscillator

- Charge q
- Current dq/dt
- Inductance L
- Resistance R
- Voltage amplitude v_0
- Oscillator frequency ω

